

COMPUTER SYSTEMS MAINTENANCE

nuts and bolts

by Robert Riggs

 In an era of exciting new developments in computer systems, data communications, and sophisticated new hardware and software, it isn't surprising that very little is said or written concerning the maintenance of 2-to-10-year-old conventional batch sequential systems that have been paying the data processing bill for so many years. These are the systems that grind through the day-to-day transactions of the business such as payroll, accounts payable, accounts receivable, billing, cost distribution, etc. In the process of being "liberated," "converted," or "recorded" several times in the past few years as we have shuffled them from one computer to another, the programs have been changed, improved, and extended to the point that they are as sensitive to the touch as a defective bomb fuse—one false move and the whole system blows. Systems analysts, programmers, and particularly the manager of systems and programming are rightfully concerned when approaching these systems. They are usually of first generation design running in emulation mode, or some equally unsatisfactory approach, on third generation computers. The manager of the computer operations is even more concerned over the condition of these systems because he never knows whether or not he can meet the production schedule. They are the bane of his existence.

This paper will discuss what can or should be done to control and improve the maintenance of these systems.

Before proceeding it may be well to describe systems maintenance in the context used here. Systems maintenance, as I use the term, is the activity associated with keeping operational computer systems continuously in tune with the requirements of users, data processing operations, associated clerical functions, and external demands from governmental and other agencies.

Usually, systems that have been around for a while constitute a fairly large segment of computer room produc-

tion. Maintenance of these systems represents a very significant demand on the systems and programming resources—possibly as high as 40% to 60% for most companies who have had a computer systems effort for a number of years. Why, under these circumstances, do we hear so little about systems maintenance? One reason is that few companies recognize the existence of systems maintenance through formal organization. Organization planners for data processing, systems design, and programming will go to any lengths to avoid the grim truth. The usual argument heard whispered at professional associations and other gatherings is that systems and programming personnel would quit or



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otherwise refuse to work in "maintenance." This attitude casts a second class citizen connotation over the systems maintenance function. So management drives maintenance underground. They will call it by many names but never identify the work for what it really is. They will organize their systems and programming groups functionally, or by product line, or by division or group, but rarely establish an organization component designated "systems maintenance." Yet any systems analyst or programmer knows whether he is working in maintenance or new systems development. No one who knows the business is fooled by calling maintenance by another name. I subscribe to the thesis that it is far better to delineate the systems maintenance function and properly staff it than to drive it underground by acting as though it doesn't exist. Not all systems analysts and programmers insist on working on the sophisticated new systems, though admittedly a great many do. The answer is to sort out the personnel between those who do and those who don't. Often the best approach is direct—ask a programmer which type of work he prefers. Careful observation of the programmer at work will sometimes provide a clue to who will gain more job satisfaction in maintenance versus original programming.

The systems and programming activities associated with maintenance can be exciting and challenging. Certainly these computer systems are dynamic. They change and they are constantly evolving. A good maintenance programmer is a highly skilled professional. He is a brilliant diagnostician and responds enthusiastically to the "factory floor" crisis in the computer room 24 hours a day, seven days a week. The payroll is due out tomorrow and the system hangs on the seventeenth program in a 20-program string. The problem may or may not be in program 17. But it is the maintenance programmer's job to find out exactly what the problem is, often among a maze of thousands of computer instructions. It may be a unique condition of input, which suddenly appears after five years of daily operation. Or it may be a hardware failure or a combination of both. Nearly all major industrial corporations are dependent upon their computers for support of the main stream of corporate record keeping. In these circumstances, good maintenance programming support is indispensable—a vital link between the computer and the customer or user or management or all three.

Maintenance programming can be more varied, more professionally demanding, and more important than any other programming assignment. Also, the relatively short time for any given assignment provides a fast turnaround to begin a new task, so the satisfaction and recognition of finishing a job and doing it well comes often.

the systems maintenance function

There are three major steps required to establish a systems maintenance function: (1) completely document all systems and programs currently in production, (2) establish a log of maintenance activity, and (3) organize. I mention good documentation as the first requirement because of its overriding importance. If you run a sloppy systems and programming shop, you might as well forget the organization and management of a systems maintenance group and just hope that the authors stay around. In fact, a very high percentage of production programs must be maintained by a programmer other than the original author. Failure to complete good documentation puts you in a vicious, never ending circle of crises.

The next step which can be taken concurrently with cleaning up the documentation is to establish a main-

tenance log. The requirements for system maintenance support are directly related to the number of requests for changes received by the system and programming group. The number of requests and the time required to service each is closely related to the types of systems in production, the number of programs in the library, the condition of documentation, and the quality of the programming staff. The maintenance log is the only really effective means of finding out what is happening. Each request for maintenance work should be logged in and show: who requested the work; a definitive statement of the problem; an estimate of the number of man-hours or days required to make the change; the name of the programmer assigned; date the request was received; date of completion; and, actual man-hours or days spent on the job. Gathering this type of information over a representative period of time will begin to show what is happening to the resources of the systems and programming group. This log of systems maintenance activity can be the basis for organization planning, allocation of manpower, and day-to-day control of computer systems maintenance work. Since maintenance of existing computer systems is a vitally important responsibility of the computer systems group, an understanding of what is really happening is essential to the effective management of the function. When the basis for analysis has been established and the systems group organized to effectively handle maintenance, the systems and programming manager will have at least part of the answer to the perennial question from his management of what all those analysts and programmers are doing.

organization

The organization alignment of the systems maintenance group could fit into the over-all edp organization with the manager of systems maintenance as one of four key managers in the total edp function.

A brief review of the functions of the three maintenance groups may be of interest. Contrary to popular opinion there is a good deal of systems analysis and evaluation work required of a well-managed maintenance group. The customer or client does not always know exactly what he wants, and would not know how to go about getting it if he did. The systems and evaluation group, within systems maintenance, should work with the user to determine exactly what is wanted, and, based on a knowledge of computer systems in general and the application involved in particular, devise the most economical and practical approach to satisfy the request. There is also a need to do some limited economic evaluation of requests for system changes. In some cases it would be far more economical and sensible to obtain the information manually or by some other means. The systems maintenance group must be constantly on guard against the tendency to dissipate systems and programming resources by frittering away manpower and computer time doing things by computer that should be done some other way. Thus the systems analysis and evaluation group plays an important role in the effective utilization of system resources. Also, this group would formalize specifications for systems maintenance requests. Failure to insist on good specifications covering systems maintenance requests has cost many companies significant amounts of money as well as wasted manpower. Without good specifications agreed to by the user, the job will take much more time to complete, because the programmer will start over or patch what he has already done as the specifications "evolve." Specifications should not be allowed to evolve. They should be established precisely and formally, and then, and only then, should work begin to program the change.

In order to control the flow of work, meet commitments, and provide excellent service to all clients or users of edp

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systems, the manager of systems maintenance must have good statistics and good library control. He needs statistics to measure work loads and allocate manpower. He needs library control procedures to assure the quality of the systems and programming documentation, both at time of entry into the library and as systems changes are implemented. The statistics required by the statistics and library control group come from the maintenance log referred to above and from data describing the program library, such as the number of programs in production, language used, and core requirements (an indication of complexity). These kinds of data should be kept current at all times.

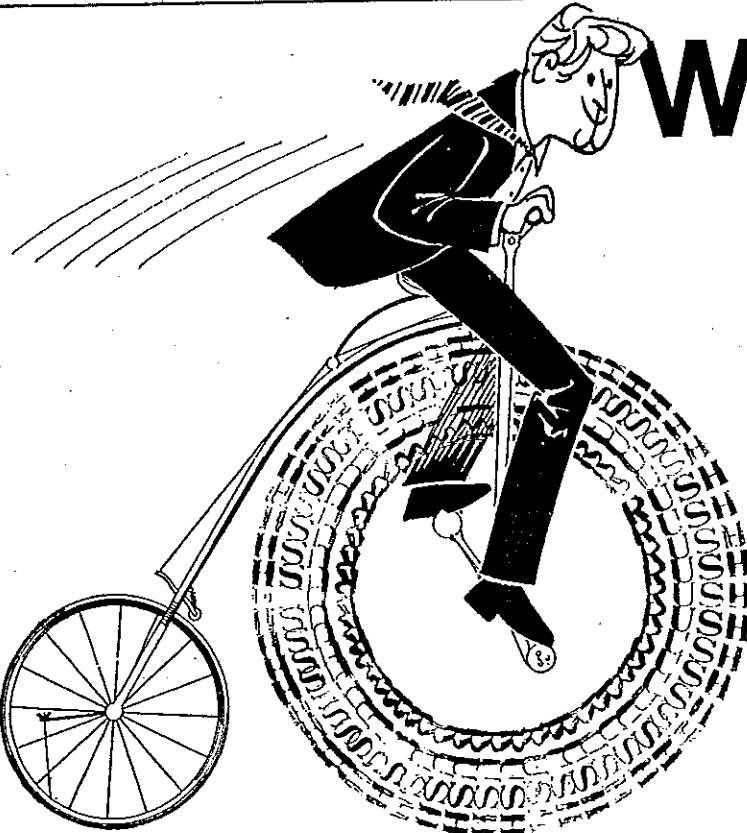
Library control procedures include: (1) a set of standards for both systems and programming documentation; (2) a procedure for securing approval from data processing operations, client user, and the maintenance manager, before a new system is transferred from new systems development to a production and maintenance status. Data processing operations should "sign off" with respect to the operational documentation and day-to-day performance of the system. The client or user should sign off that he is satisfied with the over-all effectiveness of the system and that it produces what is needed in a smooth and efficient manner. The maintenance manager should sign off to show his acceptance of both systems and programming documentation. It should be his responsibility that documentation standards are met and he should have final responsibility for accepting the new system on a maintenance basis. Unless the maintenance manager is satisfied that all requirements for transfer to a maintenance status have been met, responsibility for the system should remain with the new systems development staff.

The programming group within systems maintenance carries the actual programming responsibility for specifications established by the systems analysis and evaluation group. These are the versatile, flexible "pros" who know how to make computers hum. They are experienced in several languages, have good diagnostic abilities, and are the self-starters. They are problem solvers—the backbone of good systems maintenance.

The suggested organization is appropriate to a fairly large systems and data processing function. However, the principles of organization, separation of functions, and assignment of responsibilities apply to smaller companies.

conclusion

The executive to whom the edp function directly reports is looking for better methods of control and measurement. As these top executives increase their interest and concern, they will come to realize that management of the company's edp resources is subject to essentially the same management methods which enable them to understand, control, and measure the performance of other functional groups. Computer systems and programming activities are not a black art. As in any emerging technology, there are new words and terms being devised to enhance communications among the practitioners. The challenge lies in management looking beyond these new terms and understanding what is going on in the context of traditional management methods. When management achieves this understanding, the systems and programming manager who has failed to think through the problems of computer systems maintenance—and provide the necessary organization, manpower and professional environment to effectively manage this vital function—will find himself continuously, tragically on the defensive with top management. Current failures are contributing to a growing awareness on the part of top management that something must be done about edp systems maintenance. ■



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